

Mutation Authority as a Predictive Risk Factor: Empirical Evidence from 83 Blockchain Deployments

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Abstract—We present the Blockchain Risk Evaluation Model (BREM), a multi-domain framework that scores blockchain implementations across nine canonical dimensions organized in a Structure-Process-Persistence matrix applied to three coordination layers: Network (machine-to-machine), System State (human-to-machine), and Law (human-to-human). Applied to 83 projects spanning enterprise consortia, central bank pilots, and DeFi protocols, the model identifies a failure threshold at an overall score of 2.5 or above, achieving 92.5% sensitivity ($F1 = 0.902$) and 100% specificity among resolved projects. The central diagnostic variable is mutation authority (sm), measuring the scope of discretion any single entity retains over protocol rules, data structures, and execution environment. Of 83 projects, 76 score sm at 3 or above, and no project with sm at 2 or below has failed. The model identifies correlated risk propagation chains converging on economic nonviability, and preliminary evidence of a DeFi-specific cascade reversal ($n = 12$) that produces the same failure mechanism through a different entry point. The three false negatives are architecturally sound systems that failed commercially, suggesting the model captures systemic coordination risk more reliably than market-driven failure.

Index Terms—blockchain risk, DeFi, mutation authority, failure prediction, risk framework, systemic risk

I. INTRODUCTION

Decentralized Finance (DeFi) has produced both transformative innovation and catastrophic failure. The collapse of Terra/Luna (approximately \$40B in combined market capitalisation destroyed [13]), FTX (billions in customer funds misappropriated [14]), and Celsius (approximately \$4.7B in customer assets frozen [15]) demonstrated that DeFi risk assessment remains largely post-hoc—failures are diagnosed after capital is destroyed. Enterprise blockchain has experienced a parallel pattern: the ASX CHESS replacement consumed over A\$250M before abandonment [6], while TradeLens and Marco Polo were discontinued despite significant institutional investment, for reasons that appear, in retrospect, structurally predictable.

The central challenge is that existing risk approaches evaluate isolated dimensions—smart contract vulnerability analysis [1], transaction-ordering and MEV risk [2], or regulatory compliance—without integrating technical architecture, governance design, and legal enforceability into a unified assessment. We therefore propose an integrated framework that scores these three layers in one matrix.

This paper presents BREM (Blockchain Risk Evaluation Model), which evaluates blockchain systems across nine

canonical risk dimensions using a Structure $\times$ Process $\times$ Persistence (SPP) matrix applied at three coordination layers. We demonstrate its predictive accuracy across an 83-project dataset spanning enterprise, institutional, and DeFi deployments, and identify *mutation authority*—the scope of discretion retained over protocol rules—as the single most diagnostic risk variable.

The rest of this paper is organized as follows. Section II reviews related work. Section III presents the BREM framework and scoring methodology. Section IV describes the dataset and validation approach. Section V reports results. Section VI discusses implications and limitations. Section VII concludes.

A. Contributions

- 1) A novel 3×3 risk evaluation framework (SPP $\times$ SPP) that integrates network architecture, system governance, and legal enforceability into nine canonical risk dimensions;
- 2) An 83-project scored dataset spanning enterprise consortia, central bank pilots, and DeFi protocols, with 92.5% sensitivity and 100% specificity among resolved projects at the ≥ 2.5 threshold;
- 3) Identification of mutation authority (sm) as the defining risk variable, with empirical evidence that no project with $sm \leq 2$ has failed in the observed sample;
- 4) Preliminary identification of a DeFi-specific cascade reversal (sf $\rightarrow$ sm), observed across all five DeFi failures ($n = 12$ sub-dataset), that produces the same failure mechanism through a different causal entry point.

II. RELATED WORK

A. DeFi Risk Frameworks

DeFi risk analysis has advanced significantly in recent years. Gudgeon et al. [3] modelled the microstructure of DeFi lending protocols, analyzing interest rate dynamics, liquidity, and market efficiency. Werner et al. [4] provided a systematic classification of DeFi risks across smart contract, economic, and governance dimensions. Xu et al. [5] presented a comparative analysis of decentralized exchange designs and automated market maker mechanisms. However, these frameworks typically focus on protocol-specific risks rather than cross-domain structural patterns that predict failure across both DeFi and enterprise implementations.

B. Blockchain Failure Analysis

Post-mortem analyses of blockchain project failures have identified recurring patterns. The ASX CHESS replacement failure has been documented as an architecture-scalability problem [6]. Catalini and Gans [7] examined the economics of blockchain adoption, identifying coordination costs as a key barrier. Carson et al. [8] surveyed enterprise blockchain strategy, framing conditions under which blockchain creates business value. Wüst and Gervais [11] proposed a structured decision model for whether blockchain is appropriate, but did not quantify risk across deployed systems.

C. Governance and Centralisation Risk

The concept of protocol governance as a risk factor has been explored by Barbereau et al. [9], who characterized DeFi governance as timocratic, with tradable voting rights concentrating authority. Fritsch et al. [10] analyzed voting power distribution in DAOs. Walch [12] identified software governance as a source of centralization risk in ostensibly decentralized systems. Our contribution extends this line of work by formalizing *mutation authority*—the operational scope of discretion over protocol rules, not merely token-weighted voting power—as a scored dimension within a multi-domain framework, and demonstrating its predictive power across 83 projects spanning both enterprise and DeFi.

III. THE BREM FRAMEWORK

A. SPP $\times$ SPP Architecture

BREM applies a Structure $\times$ Process $\times$ Persistence (SPP) lens at two levels. At the macro level, the three coordination layers are SPP: Network (M2M) is the **structure** of the coordination system; System State (H2M) is the **process** operating on that structure; and Law (H2H) is the **persistence** question—whether the system survives contact with legal and institutional reality.

At the micro level, each layer is interrogated through the same SPP lens, producing nine canonical cells (Table I).

TABLE I
THE BREM 3 $\times$ 3 EVALUATION MATRIX

	Structure	Process	Persistence
Network (M2M)	na	nc	ns
System State (H2M)	se	sm	sf
Law (H2H)	ls	lr	lp

Each cell abbreviation denotes: na = Architecture, nc = Consensus, ns = Scalability, se = Execution, sm = Mutation Authority, sf = Economic Fitness, ls = Standing, lr = Remedy, lp = Liability Precision.

B. Scoring: The 0–4 Scale

Each cell is scored 0–4, where 0 represents negligible risk (protocol-constrained, no discretionary authority) and 4 represents critical risk (fully centralized control or fundamental architectural failure). The scale is calibrated to each domain:

Network measures protocol determinism (0 = independent state, no shared mutable data; 4 = shared mutable global state with sequential execution bottlenecks). **System State** measures mutation authority (0 = immutable protocol, no admin keys; 4 = database-administrator-level discretion). **Law** measures enforceability (0 = clear standing, established remedies, identifiable liable parties; 4 = non-attributable participants with no legal recourse).

Each cell is scored against domain-specific anchors defining what each score level (0–4) means for that cell. The full scoring rubric, with anchor definitions and reference examples for all nine cells, is published with the dataset.

C. Mutation Authority (sm): The Diagnostic Variable

Mutation authority measures the scope of discretion any entity retains over: (1) protocol rules and consensus parameters; (2) data structures and state management; (3) execution environment and smart contract upgradeability; (4) network parameters and access control; and (5) economic rules and fee structures.

This variable captures whether a system has committed to protocol-constrained state transition (sm = 0) or retains human override capability (sm $\geq$ 3). Unlike governance token concentration [9] or voting power distribution [10], mutation authority measures *operational* discretion: whether admin keys exist, whether contracts are upgradeable, whether parameters can be altered unilaterally. It is the architectural boundary between a blockchain and a database with distributed witnesses.

D. Scoring Extensions

The base BREM score is the unweighted mean of nine cells. Three methodological extensions refine this aggregation to address empirically observed limitations.

(1) **Asymmetric cell weighting.** The flat average treats each cell’s contribution as symmetric—a low score offsets a high score equally. However, the fail-pull (excess failure rate for above-median scores) consistently exceeds the surv-pull (excess survival rate for below-median scores) across all nine cells, i.e., no cell has an asymmetry ratio below 1.0. We apply asymmetric weights w_{up} (proportional to fail-pull) when a cell is above the risk midpoint and w_{down} (proportional to surv-pull) when below.

(2) **Domain ceiling rule.** The flat average allows a strong domain to mask a failing domain. We augment the overall threshold with a domain-level check: flag if overall $\geq$ 2.5 or any single domain average $>$ 3.0. This catches “technical zombies”—projects with strong institutional backing but technically compromised infrastructure, where the Law domain dilutes the System State signal.

(3) **Dependency decomposition.** When a project depends on N independent platforms, the mutation surface scales non-linearly: $P(\text{break}) = 1 - (1 - p)^N$. We apply a penalty $sm_{eff} = \min(4, sm_{base} + \lceil \log_2 N \rceil)$, where N counts platforms that must *interoperate for a single settlement function*, not total ecosystem platforms.

These extensions are defined here as part of the formal methodology and evaluated empirically in Section VI-C.

IV. DATASET AND METHODOLOGY

A. Dataset Composition

The dataset comprises 83 blockchain implementations:

- **Enterprise/Institutional** (n=71): Projects from financial services (securities settlement, FX, payments), trade finance, supply chain, healthcare, energy, government digital identity, and central bank digital currencies, spanning 2015–2026.
- **DeFi** (n=12): Protocol collapses (Terra/Luna, Celsius, FTX, Iron Finance, Wonderland), survivors (Uniswap V2, Aave, MakerDAO/Sky, Compound), and distressed-but-operational protocols (Axie Infinity/Ronin, Solend, Olympus DAO).

Of the 83 projects, 62 have reached definitive outcomes (*resolved*), while 21 remain in pilot, research, or early development phases (*unresolved*). Each project is documented with structured characteristics (permission model, consensus family, architecture type, consortium size) and evidence-backed scoring justifications derived from public sources.

B. Scoring Process

Each project is scored across 9 cells, each yielding a 0–4 integer score assessed against published cell-level anchors. Cell scores are averaged within rows to produce domain scores, and the overall BREM score is the mean of all nine cell scores.

Scores are derived from publicly available evidence: technical whitepapers, protocol documentation, governance forum records, on-chain contract analysis, regulatory filings, and post-mortem reports. Each score is accompanied by an evidence citation identifying the source and the specific characteristic being assessed.

The scoring process uses a structured assessment instrument with branching decision logic: for each cell, a series of diagnostic questions progressively narrows the appropriate score level, with early answers eliminating redundant downstream questions (e.g., a project confirmed as protocol-immutable need not answer questions about upgrade authority). This branching structure yields a cell-level score validated against the published anchors.

Ten anchor projects spanning the full risk spectrum (failures: ASX CRESS, Terra/Luna, TradeLens; mid-range: Project Acaia, BIS Mandala; survivors: Uniswap V2) were scored first to calibrate score-level anchors; the remaining 73 were scored against these references.

Hindsight bias mitigation. The scoring rubric assesses architectural and governance properties observable at the *design stage*, not outcomes. For example, the sm sub-questions ask whether admin keys exist, whether contracts are upgradeable, and whether governance can alter protocol parameters—all verifiable from protocol documentation before any failure occurs. To further test for outcome bias, 10 projects were scored blind to their current status by a second assessor; concordance was 89% within ± 0.5 points per cell.

C. Statistical Validation

Internal consistency was validated through Spearman rank correlation analysis between architecturally linked cells. The correlation between Architecture (na) and Scalability (ns) is $\rho = 0.606$ ($p < 0.001$), and between Scalability (ns) and Economic Fitness (sf) is $\rho = 0.566$ ($p < 0.001$), consistent with the causal chain predicted by the Architecture Death Spiral cascade (Section V-C). These moderate-to-strong correlations are expected: while the cascade model predicts directional influence, each cell also captures independent risk variation.

The status-to-score gradient provides external validation: not-failed systems average 2.12 ($\sigma = 0.45$) across all nine cells, while confirmed failures average 3.04 ($\sigma = 0.38$)—a 0.92-point separation (Cohen’s $d = 2.19$, very large effect). A Mann-Whitney U test confirms the distributions are significantly different ($p < 0.001$).

Inter-rater reliability, measured on a 10-project subsample scored by two independent assessors, yielded Cohen’s $\kappa = 0.81$ (substantial agreement) at the cell level, rising to $\kappa = 0.91$ at the domain level.

V. RESULTS

A. Failure Threshold

The 2.5 threshold was selected as the midpoint of the 0–4 scale and validated empirically: an ROC analysis across integer and half-integer thresholds from 1.5 to 3.5 identifies ≥ 2.5 as the F1-maximizing cutoff for the full dataset (F1 = 0.902), with the resolved-only subset achieving F1 = 0.961 at the same point. The threshold also corresponds to the natural separation between the not-failed mean (2.12, $\sigma = 0.45$) and the failed mean (3.04, $\sigma = 0.38$) (Table II).

TABLE II
THRESHOLD PERFORMANCE AT OVERALL SCORE ≥ 2.5

Metric	Full (n=83)	Resolved (n=62)
Sensitivity	92.5%	92.5%
Specificity	88.4%	100.0%
Precision	88.1%	100.0%
F1 Score	0.902	0.961

Among the 62 resolved projects, specificity is 100%: every project above the threshold that has reached a definitive outcome has failed or entered sustained distress. The five “false positives” in the full dataset are all unresolved projects in pilot or development phases; these represent predictions that will provide out-of-sample validation as projects mature. None has reached a definitive outcome, and all exhibit structural risk profiles comparable to confirmed failures.

1) *False Negatives: The Commercialisation Gap:* The three false negatives—HSBC FX Everywhere (2.33), NASDAQ Linq (2.33), and SETL (2.33)—share a distinctive cell-level pattern: all three have elevated System State domains (sm ≥ 3 and sf ≥ 3) but low Network risk (na ≤ 2) and low Law risk (≤ 2.0 domain average). They are architecturally sound platforms that failed on commercialisation—zero or near-zero

adoption despite regulatory approval and functional technology.

The averaging mechanism obscures the signal. In each case, the System State domain (3.0–3.33) flags genuine risk, but strong Network and Law scores dilute the overall below 2.5. The domain ceiling rule (Section III-D) catches two of the three: HSBC FX Everywhere and NASDAQ Linq both have System State > 3.0 and would be flagged under the dual threshold. SETL sits at the boundary (System State = 3.0 exactly), making it the single project in the dataset that escapes all three threshold rules.

This clarifies BREM’s scope: it identifies *systemic coordination risk* attributable to design choices, not *market adoption risk*. All three false negatives had structural capacity to operate but failed commercially—a mode outside the model’s boundary. The $sm + sf$ co-elevation is not coincidental: governance discretion ($sm \geq 3$) correlates with enterprise platforms retaining operational flexibility at the cost of protocol commitment, signalling economic models not yet validated by real usage ($sf \geq 3$).

B. Mutation Authority Distribution

Table III presents the distribution of mutation authority scores.

TABLE III
MUTATION AUTHORITY (SM) DISTRIBUTION (N=83)

sm	Count	%	Failed	Failure Rate
0	1	1.2%	0	0%
1	0	0%	—	—
2	6	7.2%	0	0%
3	42	50.6%	15	35.7%
4	34	41.0%	25	73.5%

The single $sm = 0$ project is Uniswap V2, a permissionless DeFi protocol with immutable core contracts, no admin keys, and no upgrade mechanism. Zero enterprise projects score below $sm = 2$. No project with $sm \leq 2$ has failed in the observed sample: the failure rate at $sm = 3$ is 35.7%, rising to 73.5% at $sm = 4$, while projects with bounded or absent mutation authority show a 0% failure rate ($n = 7$).

A potential confound is that sm may simply proxy for “is this an enterprise project,” since enterprise deployments require regulatory mechanisms that guarantee $sm \geq 2$. Two observations counter this interpretation. First, Terra/Luna—a permissionless DeFi protocol—scores $sm = 3$ and failed catastrophically; its governance discretion was exercised to mint billions of LUNA during the UST depeg, demonstrating that mutation authority destroys permissionless systems through the same mechanism. Second, the $sm = 3$ failure rate (35.7%) applies across *both* segments: enterprise projects at $sm = 3$ fail at comparable rates to DeFi projects at $sm = 3$. The variable measures a structural property—retained discretion over protocol rules—that operates independently of the enterprise/DeFi distinction.

C. Failure Cascades

Cell-level analysis identifies three correlated risk propagation chains (Table IV).

TABLE IV
RISK PROPAGATION CHAINS AND SUPPORTING EVIDENCE

Cascade	Path	ρ	p
Architecture $\rightarrow$ Scaling	na $\rightarrow$ ns	0.606	<0.001
Scaling $\rightarrow$ Economics	ns $\rightarrow$ sf	0.566	<0.001
Mutation $\rightarrow$ Execution	sm $\rightarrow$ se	0.462	<0.001
Execution $\rightarrow$ Standing	se $\rightarrow$ ls	0.489	<0.001

The strongest chain runs na $\rightarrow$ ns $\rightarrow$ sf: architecture constrains scalability, which prevents economic viability. Economic fitness (sf) shows the largest effect size between failed and not-failed projects (Cohen’s $d = 2.77$), followed by execution (se, $d = 1.72$) and scalability (ns, $d = 1.29$). This suggests that most failure pathways converge on economic nonviability, regardless of which domain initiates the cascade.

D. DeFi Cascade Reversal

Preliminary analysis of the DeFi sub-dataset ($n = 12$) suggests that DeFi failures follow the same propagation chains but through reversed entry points (Table V). Given the small sample size, these findings should be treated as strongly correlated observations rather than definitive causal laws; however, the pattern is consistent across all five DeFi failures.

TABLE V
ENTERPRISE VS. DeFi CASCADE COMPARISON

	Enterprise	DeFi
Primary cascade	na $\rightarrow$ ns $\rightarrow$ sf	sf $\rightarrow$ sm
Entry domain	Network	System State
Mechanism	Architecture constrains scalability	Economics expose governance
Law domain	Low risk (regulated)	High risk (no entity)
sm role	Cause	Effect

In enterprise deployments, mutation authority *causes* architectural constraints: retained governance discretion selects permissioned, centralized designs. In DeFi, economic failure *exposes* mutation authority: unsustainable tokenomics create pressure that governance discretion is then used to address, often destructively. Terra/Luna exemplifies this: the UST depeg (sf failure) exposed governance discretion that was exercised through minting billions of LUNA, accelerating the death spiral.

The DeFi sub-dataset produces separation with zero overlap: all 5 failures score ≥ 2.67 overall; all 7 non-failed projects score ≤ 2.33 . The 0.34-point gap between the highest non-failed DeFi project (Olympus DAO, 2.33) and the lowest failed (Terra/Luna, 2.67) straddles the 2.5 threshold.

E. Mutation Authority as Structural Divider

The most striking distributional finding is not domain-level averages but the sm variable itself. All 40 failures have $sm \geq 3$;

all 7 projects with $sm \leq 2$ have survived ($\chi^2 = 5.16$, $p = 0.023$). Enterprise projects average $sm = 3.42$ compared to DeFi’s $sm = 2.58$, reflecting institutional requirements for discretionary override. The enterprise consensus mechanism score ($nc = 2.89$) is also notably higher than DeFi’s ($nc = 1.83$), consistent with the observation that enterprise deployments select centralized consensus to satisfy governance constraints.

F. Illustrative Cases

Terra/Luna (2.67, Failed): the UST depeg exposed governance discretion ($sm = 3$); the Terra team minted billions of LUNA attempting to restore the peg, accelerating the death spiral (\$40B destroyed). **Uniswap V2** (1.22, Survivor): the sole $sm = 0$ project—immutable contracts, no admin keys, no upgrade mechanism. **ASX CHESS** (3.78, Failed): shared mutable global state triggered the $na \rightarrow ns \rightarrow sf$ cascade; the system could not handle concurrent clearing volumes after \$250M and seven years of development.

VI. DISCUSSION

A. The Institutional Incompatibility Proposition

The empirical pattern supports a proposition: *institutional blockchain deployments converge toward mutable governance structures, and therefore away from blockchain’s defining property of protocol-constrained state transition.*

This convergence is driven by structural requirements: regulatory compliance mandates the ability to freeze assets, reverse transactions, and enforce sanctions; governance structures require discretionary override; operational processes assume mutable records. These requirements collectively guarantee $sm \geq 2$ for any institutional deployment. The result is a system that uses blockchain vocabulary and infrastructure while preserving every discretionary mechanism that blockchain was designed to remove—what we term a “database with distributed witnesses.”

B. Implications for DeFi Resilience

The BREM matrix is a practical instrument: an analyst can score any existing or prospective protocol across nine cells using the published rubric in under an hour. Applications include pre-launch predictive scoring against the 2.5 threshold; continuous sm monitoring as an early-warning indicator of governance centralization risk (protocols with $sm \geq 3$ are structurally exposed to the $sf \rightarrow sm$ cascade); stress testing using the three propagation chains to model how architectural constraints amplify economic shocks; and governance design—the empirical record that immutability is achievable (Uniswap V2, $sm = 0$) and that retained discretion is exploitable under stress (Terra/Luna, $sm = 3$) provides concrete targets for pre-deployment architectural decisions.

C. Empirical Validation of Scoring Extensions

The three scoring extensions defined in Section III-D are evaluated against the 83-project dataset.

Asymmetric weighting. When each cell is split at its median, the asymmetry ratios are: consensus ($nc: 8.2\times$),

execution ($se: 4.5\times$), architecture ($na: 2.1\times$), legal standing ($ls: 1.8\times$). The strongest asymmetries concentrate in Network and System State; Law domain cells cluster between $1.4\times$ and $1.8\times$ —still asymmetric, but less so, consistent with the observation that legal risk can independently destroy a project while legal clarity provides only partial protection. Applying asymmetric weights improves F1 from 0.902 to 0.914 at an optimal threshold of 2.7 (Table VI).

Domain ceiling rule. The dual threshold (overall ≥ 2.5 or any domain > 3.0) catches 39/40 failures versus 37/40 for the base threshold, with the same false positive count, improving F1 to 0.929.

TABLE VI
THRESHOLD RULE COMPARISON (RESOLVED, $N=62$)

Method	Sens.	Spec.	F1
Overall ≥ 2.5	92.5%	100%	0.902
Asymmetric weighted	92.5%	100%	0.914
Overall ≥ 2.5 or domain > 3.0	97.5%	100%	0.929

The two additional catches are projects with strong institutional backing but technically compromised infrastructure—the “technical zombie” pattern. HSBC FX Everywhere (overall = 2.33) has Network and System State domains averaging 3.0 while Law averages 1.0; the flat average places it below threshold, but it failed regardless. System State is the most predictive domain (Cohen’s $d = 2.53$); averaging it equally with Law ($d = 1.25$) dilutes the strongest signal.

Dependency decomposition cross-validation. Applying BREM with dependency decomposition to three current central-bank wholesale settlement pilots confirms the effect (Table VII). All three have near-identical Law domains (≈ 1.33), similar institutional backing, and central bank oversight. The entire risk differential is architectural: the single-platform project passes every threshold rule comfortably; the systemically-coupled multi-platform project sits at the domain ceiling boundary.

TABLE VII
CROSS-VALIDATION: CENTRAL BANK SETTLEMENT PILOTS

	N	sm_{eff}	Net.	Sys.	Law	Overall
Single-platform	1	3	2.33	2.33	1.33	2.00
Modular multi	7+	3–4	3.00	2.67	1.33	2.33
Systemic multi	5	4	3.00	3.00	1.33	2.44

Critically, the modular project—which isolates use cases on separate chains—scores lower than the systemic project despite having *more* total platform dependencies, because its per-function N is 2–3 rather than 5–7. This confirms that the dependency penalty scales with per-function interoperability requirements, not total platform count. A further complication is *liability asymmetry*: platform providers typically disclaim liability for third-party integrations, capping aggregate liability at nominal amounts, rendering the project’s own governance a facade that controls rules but not the infrastructure those rules depend on.

D. Threats to Validity

The 0–4 scoring scale involves expert judgment ($\kappa = 0.81$ cell-level, 0.91 domain-level). Knowledge of outcomes may influence scoring despite the design-stage rubric; blind-scoring concordance was 89% ($n = 10$). The DeFi sub-dataset ($n = 12$) is small; DeFi-specific findings should be treated as preliminary. The model does not capture temporal dynamics as governance structures evolve.

E. Future Work

Future work includes expanding the DeFi dataset, longitudinal tracking as an early-warning system, and integration with DeFi stress testing frameworks. The 21 unresolved projects provide a natural out-of-sample validation cohort as they mature.

Agentic risk assessment. The static instrument produces point-in-time scores, but risk profiles shift with use-case scope and roadmap evolution. Future work explores using the 9-cell matrix and branching instrument as a constraint structure for an AI-driven assessment agent that conducts semi-structured protocol interviews, captures contextual factors the static rubric cannot, and flags cells where plausible scope changes would shift the score by ≥ 1 point. A prototype implementation confirms feasibility.

VII. CONCLUSION

BREM provides a structured, multi-domain framework for blockchain risk evaluation with demonstrated predictive accuracy across 83 projects. The central finding—that mutation authority is the defining variable separating protocol-constrained systems from governed databases—has implications for both DeFi protocol design and enterprise blockchain strategy.

The framework achieves 92.5% sensitivity at the ≥ 2.5 threshold with 100% specificity among resolved projects, identifies correlated risk propagation chains converging on economic nonviability, and reveals a DeFi-specific cascade reversal that produces the same structural failure through a different entry point. No project with $sm \leq 2$ has failed in the observed sample, suggesting that the architectural boundary between blockchain and database is observable, measurable, and almost universally compromised.

For DeFi resilience specifically, the model provides a predictive instrument: protocols with high mutation authority are structurally vulnerable to governance exploitation under economic stress. The framework’s principal limitation—that it captures systemic coordination risk better than commercial failure—is also its principal strength: it identifies the failure mode most directly attributable to architectural design choices.

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DATA AVAILABILITY

The scored dataset (83 projects $\times$ 40 columns), scoring rubric, codebook, and per-project evidence summaries are available at: [https://github.com/\[redacted-for-review\]](https://github.com/[redacted-for-review]). We encourage independent replication and extension.

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